JAN-T-638

20 AUGUST 1948

NATIONAL MILITARY ESTABLISHMENT SPECIFICATION

TAPE, INSULATING (ELECTRICAL), LINEN-FINISH, PLAIN

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments and supersedes the following specifications:

U. S. Army	Navy Department
6–364	17T15c
31 May 1946	15 May 1946

A. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATIONS

A–1. *Specifications.*—The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

JOINT ARMY-NAVY SPECIFICATIONS

JAN–P–105—Packaging and Packing for Overseas Shipment—Boxes, Wood, Cleated, Plywood.
JAN–P–106—Packaging and Packing for Overseas Shipment—Boxes, Wood, Nailed.
JAN–B–107—Boxes, Wood, Wirebound (Overseas Type).
JAN–P–108—Packaging and Packing for Overseas Shipment—Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.
JAN–P–125—Packaging and Packing for Overseas Shipment—Barrier-Materials, Waterproof, Flexible.
JAN–P–138—Packaging and Packing for Overseas Shipment—Boxes, Wood, Fiberboard-Lined.
JAN–P–140—Packaging and Packing for Overseas Shipment—Adhesive, Water-Resistant, Case-Liner.

U. S. ARMY SPECIFICATIONS

94–40645—Marking; Exterior, Domestic and Export Shipment by Contractors.[1]
100–2 —Standard Specification for Marking Shipments by Contractors.[2]

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material.[3]

[1] Applicable only to Air Force purchases.
[2] Applicable only to Army purchases.
[3] Applicable only to Navy purchases.

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FEDERAL SPECIFICATIONS

NN–B–591 —Boxes; Wood-Cleated-Fiberboard.
NN–B–601 —Boxes; Wood-Cleated-Plywood (for Domestic Shipments).
NN–B–621 —Boxes; Wood, Nailed and Lock-Corner.
NN–B–631 —Boxes; Wood, Wire-Bound, For Domestic Shipment.
LLL–B–631—Boxes; Fiber, Corrugated (for Domestic Shipment).
LLL–B–636—Boxes; Fiber, Solid (for Domestic Shipment).

A–2. *Other publication.*—The following publication of the issue in effect on date of invitation for bids, forms a part of this specification:

BUREAU OF SUPPLIES AND ACCOUNTS PUBLICATION

Navy Shipment Marking Handbook.[4]

B. TYPE AND SIZES

B–1. *Type.*—Insulating tape covered by this specification shall be of but one type as hereinafter specified.

B–2. *Sizes.*—Tape covered by this specification shall be furnished in the following sizes (see par. H–2):

Thickness	Width
Inch	*Inches*
0.0055	½ and ¾
0.007	¼, ½, ¾, 1, and 1½
0.011	1 and 1½

C. MATERIAL AND WORKMANSHIP

C–1. *Material.*—The tape shall be made of high-grade uniformly woven cotton, free from unsightly defects, dirt, knots, lumps, and irregularities of twist.

C–2. *Workmanship.*—The workmanship shall be first class in every respect.

D. GENERAL REQUIREMENTS

D–1. The tape shall have selvage edges, a smooth even surface commonly known as linen finish, and shall be suitable in all respects for use in insulating electrical windings.

D–2. The tape shall be unbleached and, after weaving shall receive no chemical treatment whatever. The tape shall be so made that when varnished it will have a reasonably smooth surface.

D–3. The starch and other hydrochloric-acid-soluble substances, except moisture, shall not exceed 7 percent of the dried sample. (See par. F–4a.)

[4] Applicable only to Navy purchases.

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D–4. The tape shall be put up in rolls on strawboard cores; the inside diameter of the core shall be approximately ⅜-inch. Unless otherwise specified, the roll shall contain 36 linear yards of tape. (See par. H–2.)

D–5. *Conductivity (water extract).*—The cotton used shall be free of water-soluble salts and any other electrical conducting impurities and shall meet the conductivity test of paragraph F–4d.

D–6. *Marking.*—Each roll of finished tape shall have a suitable cardboard ticket attached to the selvage, printed with the following information:

TAPE, COTTON, INSULATING
Spec. No. JAN–T–638
Width—
Thickness—
Length—
Order No.—
(Manufacturers Name or Trade-Mark)

E. DETAIL REQUIREMENTS

E–1. *Tape 0.0055-inch thick.*—

E–1a. The tape shall be of 0.0055-inch nominal thickness, and shall not vary from this thickness more than 0.0008 inch.

E–1b. The width of the tape shall not vary more than 1/32 inch from the specified width.

E–1c. The tape shall conform to the requirements shown in Table I.

TABLE I.—*Physical properties of 0.0055-inch tape.*

Width (nominal)	Number of filling threads per inch (min.)	Number of warp threads (min.)	Tensile strength (min.)	Yards per pound (min.)
Inch			*Pounds*	
½	35	39	20	400
¾	35	55	25	275

E–2. *Tape 0.007-inch thick.*—

E–2a. The nominal thickness of the tape shall be 0.007 inch. The tape thickness shall not vary from the nominal more than 0.0008 inch.

E–2b. The width of the tape shall not vary from that specified more than 1/32 inch for tape ¾ inch and less in width, or 3/64 inch for tape 1 inch or more in width.

E–2c. The tape shall conform to the requirements shown in Table II.

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TABLE II.—*Physical properties of 0.007-inch tape.*

Width (nominal)	Number of filling threads per inch		Number of warp threads		Tensile strength (min.)	Yards per pound (min.)
	Maximum	Minimum	Maximum	Minimum		
Inches					*Pounds*	
¼	39	30	35	20	20	400
½	39	35	41	38	25	260
¾	39	35	62	57	35	190
1	39	35	82	74	55	140
1½	39	35	123	110	80	80

E–3. *Tape 0.011-inch thick.*—

E–3a. The nominal thickness of the tape shall be 0.011 inch. The tape thickness shall not vary from the nominal more than 0.001 inch.

E–3b. The width of the tape shall not vary more than 3/64 inch from the specified width.

E–3c. The tape shall conform to the requirements shown in Table III.

TABLE III.—*Physical properties of 0.011-inch tape.*

Width) (nominal)	Number of filling threads per inch		Number of warp threads		Tensile strength (min.)	Yards per pound (min.)
	Maximum	Minimum	Maximum	Minimum		
Inches					*Pounds*	
1	33	31	64	61	90	70
1½	33	31	94	88	120	50

F. METHODS OF SAMPLING, INSPECTION, AND TESTS

F–1. *Sampling.*—

F–1a. *Lot.*—For purposes of sampling, a lot shall consist of all rolls of tape of the same thickness offered for delivery at one time, with the restriction that the total length of tape in a lot shall not exceed 5,000 yards.

F–1b. *Sampling procedure for lot acceptance inspection (at the place of manufacture).*—The Government inspector shall select from each lot the number of sample rolls indicated in Table IV.

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TABLE IV.—*Sampling procedure for inspection.*

Number of rolls in lot	Number of sample rolls to be selected
1–5	All
6–10	5
11–15	6
16–20	7
21–40	8
41–90	9
Over 90	10

F–1c. *Sampling procedure for lot acceptance tests.*—From each lot the Government inspector shall select 5 sample rolls of tape for the tests described in paragraph F–4. These tests shall be performed by or under the supervision of the Government inspector at the place of manufacture, if suitable facilities are available; otherwise the sample rolls shall be forwarded to the Government laboratory designated by the bureau or service concerned.

F–2. *Lot acceptance inspection (at the place of manufacture).*—

F–2a. *General.*—Each of the sample rolls selected in accordance with paragraph F–1b shall be subjected to visual inspection by the Government inspector, to determine conformance with the requirements of this specification in regard to materials, workmanship, and finish, and in regard to thickness, width and thread count. If any of the samples fails to meet the requirements of this specification, the entire lot shall be rejected. A rejected lot may be resubmitted for Government inspection provided the manufacturer, after having been informed of the reasons for rejection, has inspected each roll in the lot for the deficiency noted, and has removed all nonconforming rolls.

F–2b. *Thickness.*—The tape thickness shall be determined by the use of a micrometer having faces not less than 7/16 nor more than ½ inch in diameter, or by the use of an accurate dial gage measuring device acceptable to both the contractor and the Government inspector. At least 10 readings shall be taken of thickness, and the maximum, minimum, and average values recorded. The average value shall be taken as the thickness of the tape.

F–2c. *Width.*—The tape width shall be determined by the use of a standard steel scale graduated to read within 1/64 inch. The fabric shall lie flat on a smooth surface at the time of measurement. At least 10 measurements shall be made of width, and the maximum, minimum, and average values recorded. The average value shall be taken as the width of the tape.

F–2d. *Thread count.*—The total number of warp threads shall be determined irrespective of the width of the fabric. The number of filling threads shall be determined in a length of 1 inch. An accepted thread-counting magnifying glass shall be used in making these counts. The number of warp and of filling threads shall be taken as the average of five counts on five different places on the sample.

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F–3. *Lot acceptance tests.*—The sample rolls selected in accordance with paragraph F–1c shall be subjected to each of the tests described in paragraph F–4, i.e. the test for starch content, tensile strength, yardage and conductivity. If any roll fails to meet the requirements of this specification, the entire lot shall be rejected.

F–4. *Methods of test.*—

F–4a. *Starch content.*—

F–4a(1). The following apparatus shall be used for this test: (1) beaker, (2) hot plate or Bunsen burner and stands, (3) chemical balance, (4) ventilated drying oven, and (5) weighing bottles.

F–4a(2). Two representative test specimens of the fabric, weighing from 5 to 10 gm. each, shall be weighed separately.

F–4a(3). The first specimen shall be dried in the ventilated drying oven at 100° to 110°C. (212° to 230°F.) to constant weight. The difference between the original weight and the dry weight represents the total moisture content.

F–4a(4). The second specimen shall be boiled for 15 minutes in 500 m*l.* of a 1-percent solution of hydrochloric acid—that is, a solution containing one part of chemically pure hydrochloric acid, specific gravity 1.18, in 99 parts of water. The acid shall be removed from the specimen by washing once with distilled water and then twice in boiling distilled water, allowing the specimen to remain in the boiling water each time for 10 minutes. The specimen shall be dried to constant weight. The difference between the original weight and this dry weight after boiling in hydrochloric acid represents moisture plus hydrochloric-acid-soluble substances present. From the percentage of moisture determined on the first specimen, the dry weight of the specimen before desizing shall be calculated.

F–4a(5). The percentage of hydrochloric-acid soluble substances shall be then calculated as percentage of the dry weight of the specimen before desizing, using the following formula:

$$\frac{\text{Dry weight before desizing minus dry weight after desizing} \times 100}{\text{Dry weight before desizing}} = \text{Percentage total loss in weight}$$

F–4a(6). From the calculated percentage total loss in weight subtract 3 percentage points for cotton loss and report the result obtained as hydrochloric-acid-soluble substances such as starch or other sizing.

F–4b. *Tensile strength.*—

F–4b(1). The tensile strength shall be determined on an approved type of inclination balance testing machine. The specimens shall be tested in full section. The initial length of the test specimens between the jaws shall be 3 inches and the pulling jaw shall travel at a uniform rate of 12 inches per minute. The average of 5 tests shall be taken as the tensile strength. In the case of a break considerably below the general average of the tape a second test shall be made and this test shall then be included in obtaining the final average.

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F–4b(2). The tensile strength shall be based upon the standard moisture regain of 6.5 percent. The test specimen may be tested under prevailing atmospheric conditions and corrected to 6.5 percent regain, or it may be conditioned to the standard moisture regain before test.

F–4b(3). When tested under prevailing atmospheric conditions, correction shall be made, using one of the following formulas, as applicable:

Let x = tensile strength corrected to standard moisture regain of 6.5 percent; then,

(1) For 0.0055-inch tape,

$$x = \frac{\text{Tensile strength from machine reading} \times 107}{100 + (1.08 \times \text{actual regain at time of test})}$$

(2) For 0.007-inch tape,

$$x = \frac{\text{Tensile strength from machine reading} \times 110}{100 + (1.5 \times \text{actual regain at time of test})}$$

(3) For 0.011-inch tape,

$$x = \frac{\text{Tensile strength from machine reading} \times 116}{100 + (2.5 \times \text{actual regain at time of test})}$$

F–4b(4). When specimens are to be conditioned before testing they shall be exposed for at least 4 hours to an atmosphere of 65 percent relative humidity at a temperature of 21°C. (70°F.).

F–4c. *Yardage.*—

F–4c(1). The yardage shall be determined on the dry tape. A 12-yard sample of the tape shall be accurately measured with a steel scale by taking a portion at a time and laying it flat on a smooth surface, taking care so as not to stretch the fabric. The sample shall be weighed on a chemical balance, accurate to 1 mg. From the weight of the dry sample the number of yards per pound for the fabric shall be determined. The average of two determinations shall be considered as the yardage per pound for the tape.

F–4c(2). In the case of a dispute, a check test shall be made in accordance with the foregoing, except as follows: The 12-yard sample shall be placed in a ventilated drying oven, maintained at a temperature of 105° to 110°C. (221° to 230°F.), and shall be dried to constant weight as determined by two consecutive weighings without removal from the oven[5]. These weighings shall be taken not less than 10 minutes apart and shall show a further loss of not more than 0.1 percent of the previous weight. The results of this check shall be considered final.

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[5] If desired, the "bottle method" may be used; i. e., removing the specimen from the oven in a previously weighed, glass-stoppered bottle and carefully weighing the bottle and specimen after removal from the oven. If additional heating of the specimen is required in order to obtain constant weight, the specimen and bottle shall be placed in the oven and the stopper of the bottle removed prior to shutting the door of the oven.

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F–4d. *Conductivity (water extract).*—One-half gram of the cotton tape shall be boiled for 10 minutes in a small quantity (60 to 70 *ml.*) of distilled water which has a conductivity of not more than 5 micromhos per *ml.* at 68°F. Enough distilled water shall be added to make a volume of 100 *ml.* The conductivity of this solution as measured with alternating current shall not exceed 25 micromhos per *ml.* at 68°F.

G. PACKAGING, PACKING AND MARKING FOR SHIPMENT

G–1. *Packaging.*—Tape shall be packaged, either singly or in quantities not to exceed 12 inches when stocked, in water-proof barrier-material conforming to Joint Army-Navy Specification JAN–P–125, for interior wraps, and shall be completely sealed.

G–2. *Packing.*—

G–2a. *For domestic shipment.*—Tape, packaged as described in paragraph G–1, shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to the point of delivery. When wood-cleated-fiberboard boxes, wood-cleated-plywood boxes, nailed wood boxes, wire-bound wood boxes, corrugated fiber boxes, or solid fiber boxes are used, they shall conform to the requirements of Federal Specifications NN–B–591, NN–B–601, NN–B–621, NN–B–631, LLL–B–631 and LLL–B–636, respectively. The gross weight of wood boxes shall not exceed approximately 200 pounds; of fiberboard boxes, approximately 70 pounds.

G–2b. *For overseas shipment.*—Tape, packaged as described in paragraph G–1, shall be packed in cleated-plywood boxes, nailed wood boxes, wire-bound wood boxes, fiberboard boxes, or fiberboard-lined wood boxes conforming to the requirements of Joint Army-Navy Specifications JAN–P–105, JAN–P–106, JAN–B–107, JAN–P–108, and JAN–P–138, respectively. Shipping containers shall be lined with a sealed waterproof bag, or its equivalent, made from barrier material conforming to the requirements of Joint Army-Navy Specification JAN–P–125, for case liners. The seams and closures shall be sealed with water-resistant adhesive conforming to the requirements of Joint Army-Navy Specification JAN–P–140. The gross weight of wood boxes shall not exceed approximately 200 pounds; of fiberboard boxes, approximately 70 pounds.

G–2c. *For domestic or overseas shipment.*—Where practicable, shipping containers shall be of uniform size, and shall contain the same number of items of one size. Containers shall be designed to fit the contents in a compact manner.

G–3. *Marking.*—In addition to any special marking required by the contract or order, shipments for the Army shall be in accordance with the requirements of U. S. Army Specification 100–2; for the Air Force, in accordance with the requirements of U. S. Army Specification 94–40645; and for the Navy, in accordance with the requirements of the Navy Shipment Marking Handbook.

H. NOTES

H–1. *Use.*—Insulating tape covered by this specification is intended for use as an insulation in cables and similar items.

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H–2. Requests, requisitions, schedules and contracts or orders should specify the following:

(a) Title, number, and date of this specification.
(b) Width and thickness of tape required. (See par. B–2.)
(c) Number of yards to be contained on a single roll, if not 36 yards as provided by paragraph D–4.
(d) Whether the subject commodity is to be packed for domestic or overseas shipment. (See section G–2.)

H–3. Invitations for bids should state that bids for furnishing tape differing from the requirements of this specification will be considered provided the bidder clearly describes the specific points in which the tape it is proposed to furnish differs from the requirements of this specification and provided, further, that the differences are indicated as such. When the exceptions are not clearly described and indicated as such, it will be assumed that bidders are offering tape in strict accordance with this specification.

H–4. Copies of National Military Establishment, Joint Army-Navy, and Federal specifications (required for Army and Air Force purchases) and U. S. Army specifications may be obtained as indicated in the "Index of United States Army, Joint Army-Navy and Federal Specifications and Standards." Copies of this index may be obtained from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Services within the Departments of the Army and the Air Force will obtain copies of National Military Establishment, Joint Army-Navy, Federal, and United States Army specifications through established departmental channels. Both the title and identifying symbol number should be stipulated when requesting copies.

H–5. Copies of National Military Establishment, Joint Army-Navy, and Federal specifications (required for Navy purchases), Navy Department specifications, and the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Navy Supply Center, Norfolk 11, Va. Both the title and identifying symbol number should be stipulated when requesting copies.

Notice.—When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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